DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-400-05 — Replace spool SP04 of 4"-P300-P-1019

Area	A400 Pumps	**Task**	SR Spool replacement
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	P-104A — P-104A centrifugal pump — E 116'-6" N 18'-6" EL 100'-9"		
Equipment	P-104A	**Lines**	4"-P300-P-1019, 3/4"-P300-I-7019, 3/4"-P300-SP-7028
Isometrics	ISO-4"-P300-P-1019, ISO-3/4"-P300-I-7019, ISO-3/4"-P300-SP-7028 (generated); owner's drawings ISO-P-1019		
Permits	General work permit, Hot work permit		

Reason

Wall loss below the owner's minimum found by the last survey on the discharge run to V-102.

Scope of work

1. Isolate per the isolation list; drain, vent and purge 4"-P300-P-1019 between the isolations; confirm gas-free for hot work.
2. Remove spool 4"-P300-P-1019-SP04 (12 items, 17'-8-13/16" of pipe) — cut at the field welds and break the flanged joints at its ends as listed.
3. Fabricate the replacement spool to ISO-4"-P300-P-1019 with the same items (parts list), welding to WS-01; 11 shop welds.
4. Install the spool; make 4 field welds; new gaskets and studs at flanged ends, tightened per SW-WO-5.
5. NDE per the NDE table; hydrostatic test at 420 psi for 30 minutes with the relief valves blinded (SW-TEST-1).
6. Remove blinds, restore the line-up, hand back.

Affected items: 4"-P300-P-1019/el21, 4"-P300-P-1019/p22, 4"-P300-P-1019/el23, 4"-P300-P-1019/p24, 4"-P300-P-1019/T:3/4"-P300-I-7019, 4"-P300-P-1019/p26, 4"-P300-P-1019/T:3/4"-P300-SP-7028, 4"-P300-P-1019/p28, 4"-P300-P-1019/el29, 4"-P300-P-1019/p30, 4"-P300-P-1019/el31, 4"-P300-P-1019/p32

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-1, SW-NDE-2, SW-NDE-4, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-4, SW-WO-5, SW-WO-7, SW-WPS-1, SW-WPS-2, SW-WPS-3

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1019	4"-P300-P-1019	4"-P300-P-1019	5	3 error(s), 2 warning(s) — see below

ISO-P-1019 (PDF): ../isometrics-imported/ISO-P-1019.pdf — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1019	4	error	sheet 4: CONT'D ON ISO-P-1019 SH 5 prints E 53'-6" N -11'-6" EL 120'-0" for E 53'-6" N -10'-6" EL 120'-0": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off the line's centreline
ISO-P-1019	5	error	sheet 5: CONT'D FROM ISO-P-1019 SH 4 prints E 53'-6" N -11'-6" EL 120'-0" for E 53'-6" N -10'-6" EL 120'-0": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off the line's centreline
ISO-P-1019	5	error	W22 (4"-P300-P-1019/W22) is labelled on sheet 4 but is on sheet 5, 5.3 in from the break
ISO-P-1019	3	warning	branch weld 3/4"-P300-I-7019/W1 (E 53'-6" N 3'-6" EL 119'-3 5/16", sheet 3) has no label
ISO-P-1019	4	warning	branch weld 3/4"-P300-SP-7028/W1 (E 53'-6" N 2'-0" EL 119'-3 5/16", sheet 4) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-104A	equipment	P-104A centrifugal pump		E 116'-6" N 18'-6" EL 100'-9"
4"-P300-P-1019/el21	item	ELBOW 90° R = 5/3 OD 4" P300	4"-P300-P-1019	E 53'-9-5/16" N 5'-0" EL 116'-9-5/16"
4"-P300-P-1019/p22	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N 5'-0" EL 117'-9"
4"-P300-P-1019/el23	item	ELBOW 90° R = 5/3 OD 4" P300	4"-P300-P-1019	E 53'-6" N 4'-8-11/16" EL 118'-8-11/16"
4"-P300-P-1019/p24	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N 4'-1-5/16" EL 119'-0"
4"-P300-P-1019/T:3/4"-P300-I-7019	item	TEE REDUCING 4" × 3/4" P300	4"-P300-P-1019	E 53'-6" N 3'-6" EL 119'-0"
4"-P300-P-1019/p26	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N 2'-9" EL 119'-0"
4"-P300-P-1019/T:3/4"-P300-SP-7028	item	TEE REDUCING 4" × 3/4" P300	4"-P300-P-1019	E 53'-6" N 2'-0" EL 119'-0"
4"-P300-P-1019/p28	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N -4'-1-5/16" EL 119'-0"
4"-P300-P-1019/el29	item	ELBOW 90° R = 5/3 OD 4" P300	4"-P300-P-1019	E 53'-6" N -10'-2-11/16" EL 119'-3-5/16"
4"-P300-P-1019/p30	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N -10'-6" EL 120'-0"
4"-P300-P-1019/el31	item	ELBOW 90° R = 5/3 OD 4" P300	4"-P300-P-1019	E 53'-6" N -10'-9-5/16" EL 120'-8-11/16"
4"-P300-P-1019/p32	item	PIPE 4" P300 BEVELLED ENDS	4"-P300-P-1019	E 53'-6" N -12'-1-3/4" EL 121'-0"
4"-P300-P-1019/A5	item	VALVE, GATE, WEDGE, OS&Y 3" P300	4"-P300-P-1019	E 116'-6" N 16'-6" EL 105'-2-3/16"
4"-P300-P-1019/A8	item	FLANGE WELD-NECK 3" P300 RAISED FACE	4"-P300-P-1019	E 116'-6" N 16'-6" EL 105'-11-5/8"

Tag	Kind	Description	Line	Location
4"-P300-P-1019/B16	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1019	E 53'-6" N -14'-4" EL 121'-0"
4"-P300-P-1019/B19	item	FLANGE WELD-NECK 4" P300 RAISED FACE	4"-P300-P-1019	E 53'-6" N -13'-5-1/2" EL 121'-0"
4"-P300-P-1019/W12	weld	Butt weld 4" P300 (field)	4"-P300-P-1019	E 54'-11/16" N 5'-0" EL 116'-6"
4"-P300-P-1019/W13	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 5'-0" EL 117'-11/16"
4"-P300-P-1019/W14	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 5'-0" EL 118'-5-5/16"
4"-P300-P-1019/W15	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 4'-5-5/16" EL 119'-0"
4"-P300-P-1019/W16	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 3'-9-5/16" EL 119'-0"
4"-P300-P-1019/W17	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 3'-2-11/16" EL 119'-0"
4"-P300-P-1019/W18	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 2'-3-5/16" EL 119'-0"
4"-P300-P-1019/W19	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N 20-11/16" EL 119'-0"
4"-P300-P-1019/W20	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N -9'-11-5/16" EL 119'-0"
4"-P300-P-1019/W21	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N -10'-6" EL 119'-6-11/16"
4"-P300-P-1019/W22	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N -10'-6" EL 120'-5-5/16"
4"-P300-P-1019/W23	weld	Butt weld 4" P300 (shop)	4"-P300-P-1019	E 53'-6" N -11'-11/16" EL 121'-0"
4"-P300-P-1019/W24	weld	Butt weld 4" P300 (field)	4"-P300-P-1019	E 53'-6" N -13'-2-13/16" EL 121'-0"
3/4"-P300-I-7019/W1	weld	Butt weld 0.75" P300 (field)	3/4"-P300-I-7019	E 53'-6" N 3'-6" EL 119'-3-5/16"
3/4"-P300-SP-7028/W1	weld	Butt weld 0.75" P300 (field)	3/4"-P300-SP-7028	E 53'-6" N 2'-0" EL 119'-3-5/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1019/A5	4"-P300-P-1019	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (4"-P300-P-1019/A5)	E 116'-6" N 16'-6" EL 105'-2-3/16"		
2	blind	4"-P300-P-1019/A8	4"-P300-P-1019	Fit a line blind between 4"-P300-P-1019/A5 and FLANGE WELD-NECK 3" P300 RAISED FACE (4"-P300-P-1019/A8), work side	E 116'-6" N 16'-6" EL 105'-9-1/16"		
3	valve	4"-P300-P-1019/B16	4"-P300-P-1019	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1019/B16)	E 53'-6" N -14'-4" EL 121'-0"		
4	blind	4"-P300-P-1019/B19	4"-P300-P-1019	Fit a line blind between 4"-P300-P-1019/B16 and FLANGE WELD-NECK 4" P300 RAISED FACE (4"-P300-P-1019/B19), work side	E 53'-6" N -13'-8-1/4" EL 121'-0"		

Parts

Description	Qty	Replaces
ELBOW 90° R = 5/3 OD 4" P300	4	
PIPE 4" P300 BEVELLED ENDS	6	
TEE REDUCING 4" × 3/4" P300	2	

New welds (SW-WO-4)

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-400-05-W01	4"-P300-P-1019/W12	ISO-P-1019 sh 2 W12	4"-P300-P-1019	4" P300	0.1875 in	field	WS-01	VT + RT	E 54'-11/16" N 5'-0" EL 116'-6"		
WO-400-05-W02	4"-P300-P-1019/W13	ISO-P-1019 sh 2 W13	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 5'-0" EL 117'-11/16"		
WO-400-05-W03	4"-P300-P-1019/W14	ISO-P-1019 sh 3 W14	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 5'-0" EL 118'-5-5/16"		
WO-400-05-W04	4"-P300-P-1019/W15	ISO-P-1019 sh 3 W15	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 4'-5-5/16" EL 119'-0"		
WO-400-05-W05	4"-P300-P-1019/W16	ISO-P-1019 sh 3 W16	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 3'-9-5/16" EL 119'-0"		
WO-400-05-W06	4"-P300-P-1019/W17	ISO-P-1019 sh 3 W17	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 3'-2-11/16" EL 119'-0"		
WO-400-05-W07	4"-P300-P-1019/W18	ISO-P-1019 sh 4 W18	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 2'-3-5/16" EL 119'-0"		
WO-400-05-W08	4"-P300-P-1019/W19	ISO-P-1019 sh 4 W19	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N 20-11/16" EL 119'-0"		

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-400-05-W09	4"-P300-P-1019/W20	ISO-P-1019 sh 4 W20	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N -9'-11-5/16" EL 119'-0"		
WO-400-05-W10	4"-P300-P-1019/W21	ISO-P-1019 sh 4 W21	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N -10'-6" EL 119'-6-11/16"		
WO-400-05-W11	4"-P300-P-1019/W22	ISO-P-1019 sh 5 W22	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N -10'-6" EL 120'-5-5/16"		
WO-400-05-W12	4"-P300-P-1019/W23	ISO-P-1019 sh 5 W23	4"-P300-P-1019	4" P300	0.1875 in	shop	WS-01	VT + RT	E 53'-6" N -11'-11/16" EL 121'-0"		
WO-400-05-W13	4"-P300-P-1019/W24	ISO-P-1019 sh 5 W24	4"-P300-P-1019	4" P300	0.1875 in	field	WS-01	VT + RT	E 53'-6" N -13'-2-13/16" EL 121'-0"		
WO-400-05-W14	3/4"-P300-I-7019/W1	ISO-P-1019 sh 3 (not labelled; to read I-7019/W1)	3/4"-P300-I-7019	0.75" P300	0.09375 in	field	WS-01	VT + RT	E 53'-6" N 3'-6" EL 119'-3-5/16"		
WO-400-05-W15	3/4"-P300-SP-7028/W1	ISO-P-1019 sh 4 (not labelled; to read SP-7028/W1)	3/4"-P300-SP-7028	0.75" P300	0.09375 in	field	WS-01	VT + RT	E 53'-6" N 2'-0" EL 119'-3-5/16"		

Welding specification WS-01 — Butt welds, pipe grade, wall to 1/2 in

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Scope	Pipe, fittings and weld-neck flanges of Grade 35 pipe / Grade 36 forging, wall up to 1/2 in
Joint	Single-V, 60° included, root opening 1/8 in, root face 1/16 in (SW-WD-4)
Process	Root: gas-shielded tungsten arc. Fill and cap: shielded metal arc.
Filler	Root SW-F1 wire; fill and cap SW-F2 covered electrode (matching strength, low hydrogen)
Preheat	≥ 30 °F above dew point and ≥ 60 °F, 3 in each side
Interpass	≤ 450 °F
Post-weld heat treatment	Not specified (owner to state)
Rules	SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-WD-4

NDE

Method	Extent	Welds	Basis
VT	100 % of new welds, before and after welding	WO-400-05-W01, WO-400-05-W02, WO-400-05-W03, WO-400-05-W04, WO-400-05-W05, WO-400-05-W06, WO-400-05-W07, WO-400-05-W08, WO-400-05-W09, WO-400-05-W10, WO-400-05-W11, WO-400-05-W12, WO-400-05-W13, WO-400-05-W14, WO-400-05-W15	SW-NDE-1
RT (UT where RT is impractical)	100 % of new welds (class H300, high consequence)	WO-400-05-W01, WO-400-05-W02, WO-400-05-W03, WO-400-05-W04, WO-400-05-W05, WO-400-05-W06, WO-400-05-W07, WO-400-05-W08, WO-400-05-W09, WO-400-05-W10, WO-400-05-W11, WO-400-05-W12, WO-400-05-W13, WO-400-05-W14, WO-400-05-W15	SW-NDE-2

Test — Hydrostatic test

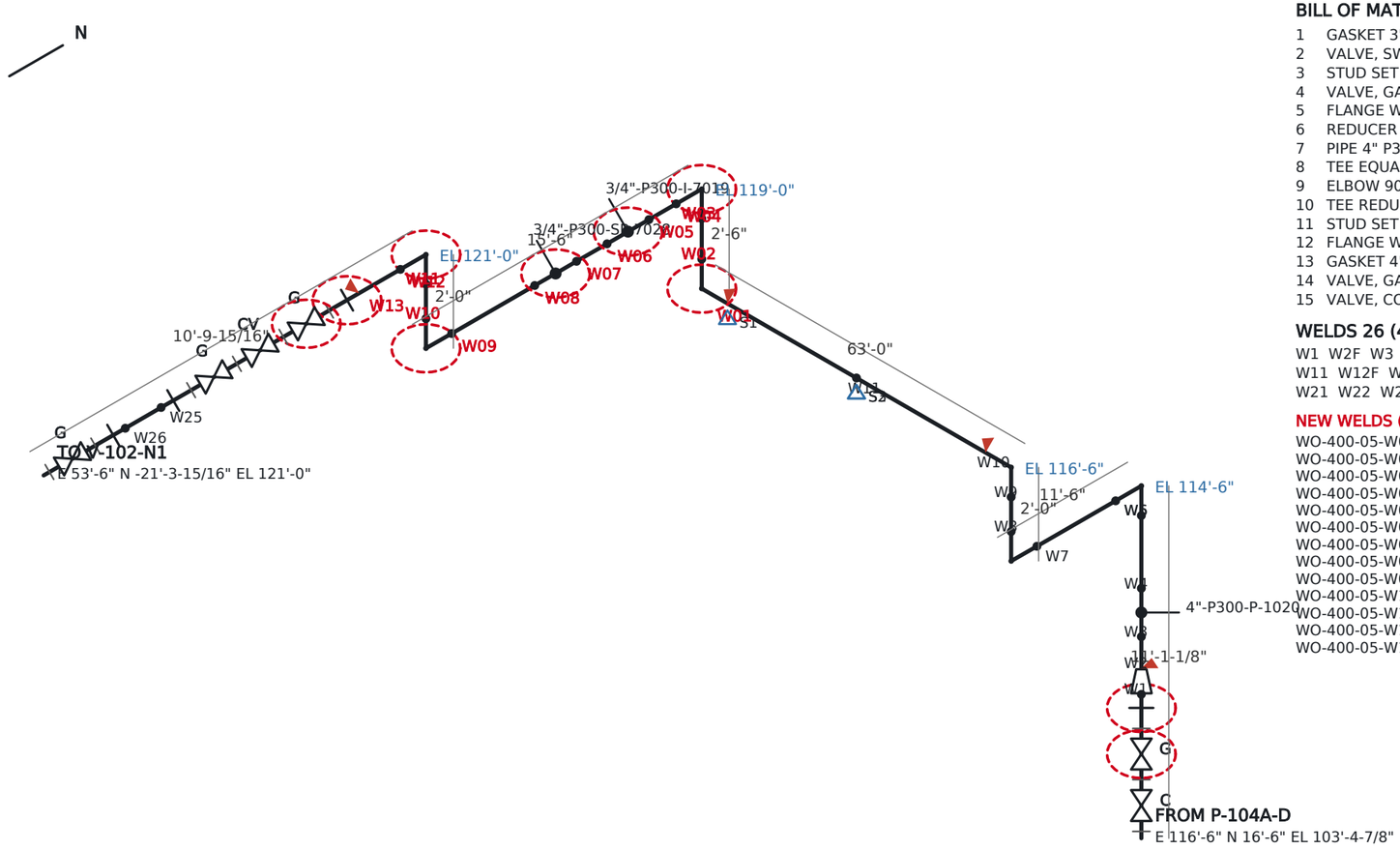
Description	Test boundary between the isolations; relief valves blinded or removed
Pressure	420 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Line drained, vented and purged; gas-free for hot work	Owner hot-work procedure	Gas test within owner limits	Gas test record, hot work permit	H	H	—				
3	Replacement materials verified against the parts list (size, class, grade, heat marking)	Parts list, SW-LC-3	All items match the line class	Material receiving record	R	R	R				
4	Welding specification WS-01 and welder identification available at the work	WS-01, SW-WPS-3	Specification on site; every welder identified	Welder list	R	R	R				
5	Fit-up of every new weld: root opening and alignment within the root face	SW-WPS-3, SW-WD-4	Root opening 1/8 in; bores aligned within 1/16 in	Fit-up record per weld	H	H	S				
6	Preheat and interpass temperatures maintained during welding	WS-01, SW-WPS-2	Preheat ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side; interpass ≤ 450 °F	Weld record per weld (welder, temperatures)	S	S	S				
7	Visual examination of every new weld	SW-NDE-1, SW-NDE-4	No crack, lack of fusion or incomplete penetration	VT report	W	W	W				
8	Volumetric examination per the NDE table; repairs re-examined	SW-NDE-2, SW-NDE-4	Acceptance per SW-NDE-4 and the owner's criteria	RT/UT reports	R	H	R				
9	Flanged ends: faces inspected, new gaskets, studs to preload	SW-NDE-5, SW-WO-5	No radial scratch; preload as stated	Bolt-up record	W	W	—				
10	Hydrostatic test at 420 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record with gauge calibration	H	H	H				
11	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
12	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1019 — Replace spool SP04 of 4"-P300-P-1019



BILL OF MATERIALS

1	GASKET 3" P300 COMPRESSED SHEET	3
2	VALVE, SWING CHECK 3" P300	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	3
4	VALVE, GATE, WEDGE, OS&Y 3" P300	1
5	FLANGE WELD-NECK 3" P300 RAISED FACE	1
6	REDUCER CONCENTRIC 4" × 3" P300	1
7	PIPE 4" P300 BEVELLED ENDS	99'-2-15/16"
8	TEE EQUAL 4" P300	1
9	ELBOW 90° R = 5/3 OD 4" P300	7
10	TEE REDUCING 4" × 3/4" P300	2
11	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	6
12	FLANGE WELD-NECK 4" P300 RAISED FACE	3
13	GASKET 4" P300 COMPRESSED SHEET	6
14	VALVE, GATE, WEDGE, OS&Y 4" P300	3
15	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	1

WELDS 26 (4 FIELD) · SUPPORTS 3 · SPOOLS 6

W1 W2F W3 W4 W5 W6 W7 W8 W9 W10F
W11 W12F W13 W14 W15 W16 W17 W18 W19 W20
W21 W22 W23 W24F W25 W26

NEW WELDS (WO-400-05)

WO-400-05-W01 replaces W12
WO-400-05-W02 replaces W13
WO-400-05-W03 replaces W14
WO-400-05-W04 replaces W15
WO-400-05-W05 replaces W16
WO-400-05-W06 replaces W17
WO-400-05-W07 replaces W18
WO-400-05-W08 replaces W19
WO-400-05-W09 replaces W20
WO-400-05-W10 replaces W21
WO-400-05-W11 replaces W22
WO-400-05-W12 replaces W23
WO-400-05-W13 replaces W24

4"-P300-P-1019

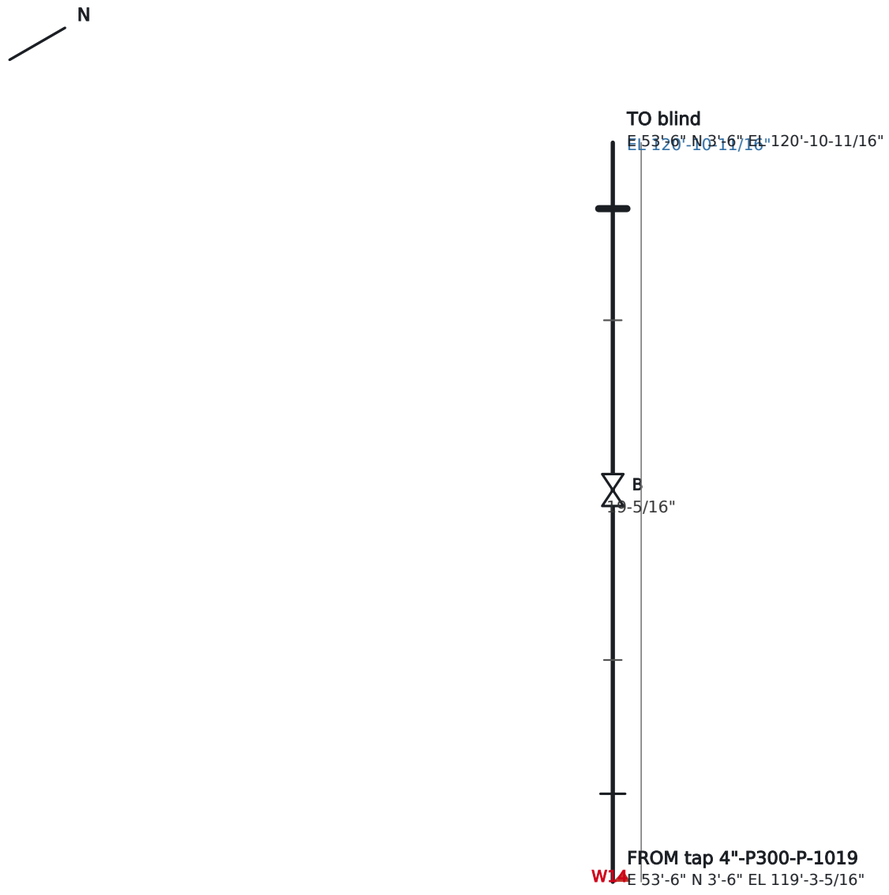
CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
P-104A-D → V-102-N1
PIPE 99'-2-15/16" · WEIGHT 1884 lb

DWG ISO-4"-P300-P-1019 REV 0 WO-400-05

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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ISOMETRIC — 3/4"-P300-I-7019 — Replace spool SP04 of 4"-P300-P-1019



BILL OF MATERIALS

1	FLANGE WELD-NECK 3/4" P300 RAISED FACE	1
2	GASKET 3/4" P300 COMPRESSED SHEET	2
3	VALVE, BALL, FLOATING 3/4" P300	1
4	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
5	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 1 (1 FIELD) · SUPPORTS 0 · SPOOLS 1

W1F

NEW WELDS (WO-400-05)

WO-400-05-W14 replaces W1

3/4"-P300-I-7019

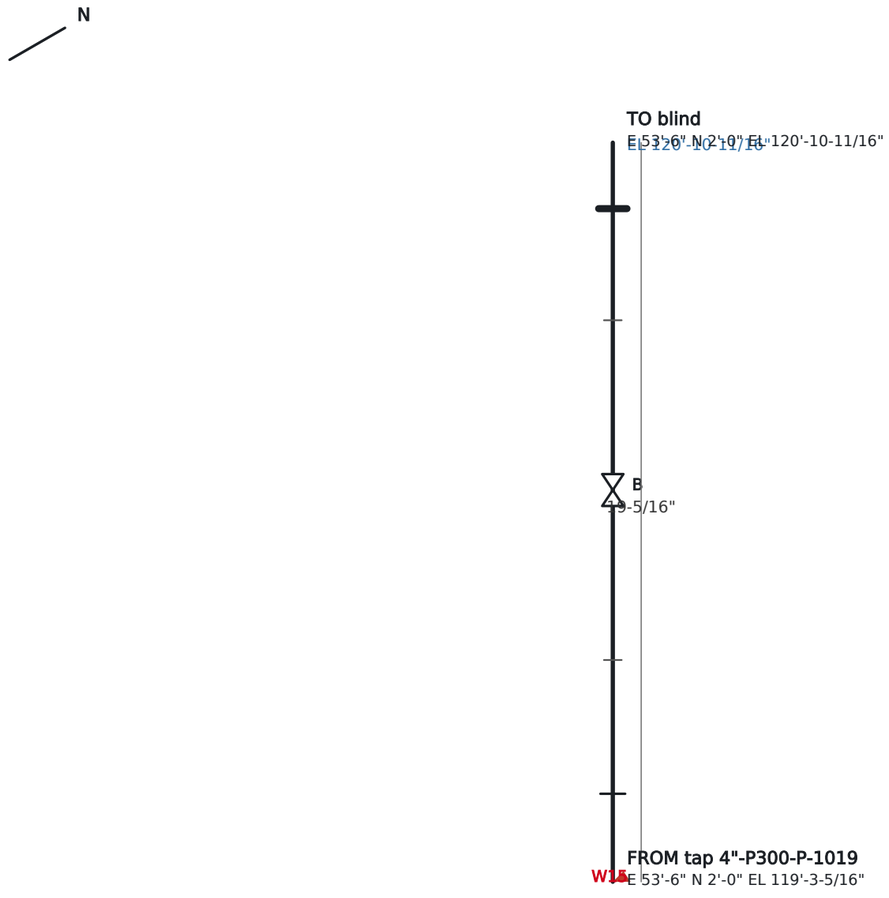
CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
tap 4"-P300-P-1019 → blind
PIPE 0.000" · WEIGHT 37 lb

DWG ISO-3/4"-P300-I-7019 REV 0 WO-400-05

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

ISOMETRIC — 3/4"-P300-SP-7028 — Replace spool SP04 of 4"-P300-P-1019



BILL OF MATERIALS

1	FLANGE WELD-NECK 3/4" P300 RAISED FACE	1
2	GASKET 3/4" P300 COMPRESSED SHEET	2
3	VALVE, BALL, FLOATING 3/4" P300	1
4	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
5	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 1 (1 FIELD) · SUPPORTS 0 · SPOOLS 1

W1F

NEW WELDS (WO-400-05)

WO-400-05-W15 replaces W1

3/4"-P300-SP-7028

CLASS H300 — Hydrocarbon process
SERVICE SP · 0.75" · P300 · DESIGN 300 psi
tap 4"-P300-P-1019 → blind
PIPE 0.000" · WEIGHT 37 lb

DWG ISO-3/4"-P300-SP-7028 REV 0 WO-400-05

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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